Discriminative Energy Component Analysis:
Optimal Commuting Measurements and Spectral Quadratic Classification

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Abstract

Principal component analysis orders directions by total variance, not by class discrimination. Eigen-Component Analysis (ECA) instead squares orthogonal projections and assigns the resulting directional energies to classes, but its precise optimization target, quantum interpretation, and limits have remained unclear. We reformulate this idea as *Discriminative Energy Component Analysis* (DECA): minimum-error state discrimination under a shared-eigenbasis, or commuting-measurement, constraint. A common unitary is provably redundant for an unrestricted positive operator-valued measure (POVM), so its only operational role is measurement compression. Under this constraint, we prove that the optimal fixed-basis decoder is hard; binary DECA has a closed-form eigensolution equal to the known Helstrom optimum; commuting multiclass ensembles are exact; and the loss relative to the optimal POVM is bounded by a joint-diagonalization residual. For noncommuting multiclass problems we give a monotone Jacobi algorithm with analytical two-dimensional updates. We also separate single-shot PVM-DECA from Spectral-DECA, which retains the magnitude of each class-difference eigenvalue for deterministic quadratic classification. Qiskit Aer simulations verify ancilla-free PVM and Naimark-dilated POVM circuits to maximum total-variation error 0.0073. In 1,100 repeated cross-validation fits on ten datasets, DECA validates the intended second-order mechanism but does not generally outperform classical baselines. These results establish a rigorous resource–accuracy boundary, not a quantum-speedup or universal-accuracy claim.

1 Introduction

Variance and discrimination are different objectives. Principal component analysis (PCA) chooses directions that explain total second-moment energy, including nuisance variation shared by all classes. A classifier instead needs directions whose responses differ across classes. Fisher's linear discriminant analysis (LDA) addresses this distinction through class means [1]; quadratic discriminant analysis (QDA), common spatial patterns, and common principal components expose complementary second-order structure [2–4]. The motivating question of this work is narrower:

> Can a single orthogonal basis be ordered and allocated by *class-conditional energy contrast*, rather than total variance, while remaining analytically tractable and physically measurable?

The original ECA preprint [5] and the later ISCAS paper [6] proposed squared orthogonal components,

$$z_j(x) = |p_j^\dagger \phi(x)|^2, \tag{1}$$

followed by class weights. This construction is linear in $z(x)$ but is a structured quadratic classifier in the encoded input $\phi(x)$. When the weights are nonnegative and sum to one across classes for every component, the class operators are a commuting POVM. This observation is the starting point here.

Quantum state discrimination already supplies the binary Helstrom measurement [7], general POVM optimization by semidefinite programming (SDP) [8], and the square-root or Pretty Good Measurement (PGM) [9]. These tools have also been applied to classical supervised classification [10–14]. Consequently, "using a density matrix" or "using Helstrom" is not a novelty claim. The unresolved issue is what remains of ECA after those known facts are imposed exactly.

Contributions. This paper makes four scoped contributions.

1. We identify ECA with a commuting-measurement constraint and prove that a shared unitary is irrelevant to an unrestricted POVM but meaningful as a compression of measurement resources.

2. We analytically eliminate the soft component-to-class decoder, derive binary and commuting-multiclass exactness, prove a computable off-diagonal residual bound, and expose the unavoidable $K > d$ output limitation of a d-dimensional PVM.

3. We introduce a monotone multiclass Jacobi-DECA procedure and separate the single-shot PVM rule from a spectral observable rule that preserves the magnitudes of discriminative eigenvalues.

4. We provide tested classical, SDP, and Qiskit implementations; validate all stated identities numerically; and report both positive and negative results over controlled ensembles and public datasets.

We do *not* claim quantum speedup, privacy, hardware power or latency advantages, or universal predictive superiority.

2 Related Work and Positioning

2.1 Discriminative second-order representations

LDA finds a mean-separating subspace relative to within-class scatter [1]. QDA assigns each class an unrestricted covariance model, while regularized discriminant analysis interpolates between linear and quadratic rules [2]. Common spatial patterns diagonalize two covariance matrices to contrast class-specific variance and are widely used for EEG decoding [4]. Flury's common principal component model assumes several covariance matrices share eigenvectors [3]. Approximate joint diagonalization and Jacobi plane rotations provide computational tools when exact commutation fails [15, 16].

DECA is closest to these methods statistically: its sufficient statistics are class-conditional means of rank-one energy operators. It differs from unrestricted QDA by forcing class quadratic forms to share a basis, and from PCA by ordering directions through class contrast rather than pooled variance. The shared-basis assumption is a bias that can either reduce estimation variance or cause severe underfitting.

2.2 Quantum state discrimination classifiers

For a known binary ensemble, the Helstrom measurement minimizes single-copy Bayes error [7]. Multiclass minimum-error discrimination is a convex SDP [8]; PGM is a closed-form feasible alternative [9]. Sergioli *et al.* constructed class density patterns and a Helstrom quantum centroid classifier [10]. Giuntini *et al.* extended state-discrimination ideas to supervised multiclass classification [11]. Lloyd *et al.* studied trainable embeddings whose trace distance is paired with a Helstrom measurement [12]. Later work developed PGM and SDP classifiers [13] and efficient simulations of multicopy Helstrom rules [14].

Our binary solution is an independent reduction of the ECA objective to the known Helstrom result, not a new state-discrimination theorem. The original content lies in the constrained-measurement formulation, hard-decoder elimination, multiclass residual analysis, Jacobi objective, spectral/PVM distinction, and end-to-end resource study.

3 Problem Formulation

Let an input be encoded as a unit vector $\phi(x) \in \mathbb{C}^d$, with pure-state operator

$$\rho_x = \phi(x)\phi(x)^\dagger, \qquad \mathrm{Tr}(\rho_x) = 1. \tag{2}$$

For class $c \in \{1, \dots, K\}$, define

$$\rho_c = \mathbb{E}[\rho_x \mid y = c], \qquad A_c = \pi_c \rho_c, \tag{3}$$

where π_c is the class prior. Thus $A_c \succeq 0$ and $\sum_c \mathrm{Tr}(A_c) = 1$.

The general minimum-error measurement is

$$S^\star_{\mathrm{POVM}} = \max_{\substack{E_c \succeq 0 \\ \sum_c E_c = I}} \sum_{c=1}^{K} \mathrm{Tr}(A_c E_c). \tag{4}$$

Its dual is

$$S^\star_{\mathrm{POVM}} = \min_Y \mathrm{Tr}(Y) \quad \text{s.t.} \quad Y \succeq A_c \quad \forall c. \tag{5}$$

We use Equation (4) as a numerical oracle, not as a contribution.

Proposition 1 (Shared-unitary redundancy)**.** *For every unitary U,*

$$S^\star_{\mathrm{POVM}}(\{UA_cU^\dagger\}) = S^\star_{\mathrm{POVM}}(\{A_c\}).$$

Proof. The map $E_c \mapsto UE_cU^\dagger$ is a bijection on feasible POVMs and preserves every trace term by cyclicity. □

Proposition 1 prevents a common conceptual error: a learned unitary cannot create information if the subsequent measurement is unrestricted. It matters only when the measurement family, circuit depth, or number of outcomes is restricted.

4 DECA as an Optimal Commuting Measurement

Restrict all effects to a shared unitary basis $P = [p_1, \dots, p_d]$:

$$E_c = P\,\mathrm{diag}(\ell_{1c}, \dots, \ell_{dc})P^\dagger, \quad \ell_{jc} \geq 0, \quad \sum_c \ell_{jc} = 1. \tag{6}$$

Let $a_{jc}(P) = p_j^\dagger A_c p_j$. The success is

$$S(P, L) = \sum_{j,c} \ell_{jc} a_{jc}(P). \tag{7}$$

Theorem 2 (Hard-decoder optimality)**.** *For fixed P, an optimal decoder exists at a vertex of every row simplex:*

$$\ell^\star_{jc} = \mathbf{1}\left[c = \arg\max_r a_{jr}(P)\right],$$
$$S_{\mathrm{DECA}}(P) = \sum_j \max_c a_{jc}(P). \tag{8}$$

Proof. Each row $\ell_{j:}$ appears in an independent linear program over the probability simplex. A maximizing vertex assigns all mass to a largest coefficient. $\square$

Thus soft weights are unnecessary for the minimum-error objective. The optimal commuting POVM can be chosen as a projective measurement followed by a classical outcome-to-class map.

4.1 Binary closed form

Let $\Delta = A_1 - A_2$. For every P,

$$S_{\mathrm{DECA}}(P) = \frac{1}{2}\left[1 + \left\|\mathrm{diag}(P^\dagger \Delta P)\right\|_1\right]. \tag{9}$$

Theorem 3 (Binary DECA equals Helstrom)**.** *If P is an eigenbasis of Δ, then*

$$S^\star_{\mathrm{DECA}} = \frac{1}{2}(1 + \|\Delta\|_1) = S^\star_{\mathrm{POVM}}. \tag{10}$$

Outcome j is assigned according to the sign of $\lambda_j(\Delta)$.

Proof. By the Schur–Horn theorem, $\mathrm{diag}(P^\dagger \Delta P) \prec \lambda(\Delta)$ [17]. The ℓ_1 norm is convex and symmetric, hence $\|\,\mathrm{diag}(P^\dagger \Delta P)\|_1 \le \|\lambda(\Delta)\|_1$. An eigenbasis attains equality. Substitution into Equation (9) yields the known Helstrom formula. $\square$

The magnitude $|\lambda_j|$ is the contribution of component j to total trace-distance discrimination. This gives a principled low-rank ordering: retain the smallest r satisfying $\sum_{j\le r} |\lambda|_{(j)} / \sum_j |\lambda_j| \ge \eta$.

4.2 Multiclass exactness and approximation

Theorem 4 (Commuting multiclass exactness)**.** *If $[A_c, A_r] = 0$ for all c, r, then a common eigenbasis P satisfies $S_{\mathrm{DECA}}(P) = S^\star_{\mathrm{POVM}}$.*

Proof. In the common eigenbasis, only the diagonal entries of any feasible POVM contribute to Equation (4). Those entries form a stochastic decoder, whose optimum is the hard decoder in Theorem 2. $\square$

For a general basis define

$$P^\dagger A_c P = D_c + O_c, \quad D_c = \mathrm{diag}(P^\dagger A_c P), \tag{11}$$

and

$$R_{\mathrm{off}}(P) = \left(\sum_c \|O_c\|_F^2\right)^{1/2}. \tag{12}$$

Theorem 5 (POVM gap bound)**.** *For every unitary P,*

$$0 \le S^\star_{\mathrm{POVM}} - S_{\mathrm{DECA}}(P) \le \sqrt{d}\, R_{\mathrm{off}}(P). \tag{13}$$

Proof. Write an optimal POVM in the P basis as $\{F_c\}$. Its diagonal part is a feasible soft decoder, no better than the fixed-basis hard optimum. The remaining contribution is at most $\sum_c |\operatorname{Tr}(F_c O_c)|$. Cauchy–Schwarz and $0 \preceq F_c \preceq I$ give

$$\begin{aligned}\sum_c |\operatorname{Tr}(F_c O_c)| &\le \left(\sum_c \|F_c\|_F^2\right)^{1/2} R_{\mathrm{off}} \\ &\le \sqrt{\sum_c \operatorname{Tr}(F_c) R_{\mathrm{off}}} = \sqrt{d}\, R_{\mathrm{off}}.\end{aligned}$$

$\square$

The bound is intentionally one-sided and can be loose. A commutator norm is also a useful diagnostic, but our experiments show it need not be monotone in the actual oracle gap.

Proposition 6 ($K > d$ PVM capacity)**.** *A d-dimensional PVM followed by a hard decoder has at most d nonzero class effects. If $K > d$, at least $K - d$ classes lack an independent nonzero effect.*

Proof. Nonzero orthogonal projectors have mutually orthogonal supports, whose ranks sum to at most d. Coarse-graining outcomes cannot increase their count. $\square$

5 Spectral-DECA: Retaining Difference Magnitudes

The single-shot objective and deterministic classification are not interchangeable. In the binary eigenbasis $\Delta = P\,\mathrm{diag}(\lambda)P^\dagger$, exact-probability PVM inference uses

$$g_{\mathrm{PVM}}(x) = \sum_j \mathrm{sign}(\lambda_j) |p_j^\dagger \phi(x)|^2, \tag{14}$$

whereas the class-operator difference is

$$\boxed{g_{\mathrm{spec}}(x) = \operatorname{Tr}(\Delta \rho_x) = \sum_j \lambda_j |p_j^\dagger \phi(x)|^2.} \tag{15}$$

We call Equation (15) Spectral-DECA. It is an analytical structured quadratic discriminant that preserves both the sign and magnitude of class difference. It is not a new single-shot measurement optimum.

On a quantum device, apply the same $P^\dagger$, measure repeatedly in the computational basis, and weight outcome J by λ_J. If $r_\lambda = \lambda_{\max} - \lambda_{\min}$, Hoeffding's inequality [18] gives

$$\Pr(|\hat{g} - g_{\mathrm{spec}}| \ge \epsilon) \le 2\exp\left(-\frac{2N\epsilon^2}{r_\lambda^2}\right). \tag{16}$$

Spectral-DECA therefore uses no extra basis or ancilla, but exchanges a single-shot decision for repeated expectation estimation.

Algorithm 1 Jacobi-DECA

Require: Weighted operators $\{A_c\}_{c=1}^{K}$; initial bases $\{P_s^{(0)}\}$; tolerance ε; sweep budget T

for each initial basis $P_s^{(0)}$ **do**
  for $t = 0, \dots, T-1$ **do**
    $z_j \leftarrow \arg\max_c p_j^\dagger A_c p_j$ for all j
    for all pairs $j < k$ with $z_j \neq z_k$ **do**
      $B \leftarrow [p_j, p_k]^\dagger (A_{z_j} - A_{z_k})[p_j, p_k]$
      replace (p_j, p_k) by the eigenbasis of B, largest eigenvector first
    recompute $F(P)$; stop if relative gain $\leq \varepsilon$
return the candidate with largest $F(P)$ and its hard decoder

For $K > 2$, use the diagonal common-basis approximation

$$\widehat{A}_c(P) = P\,\mathrm{diag}(P^\dagger A_c P)P^\dagger, \qquad q_c(x) = \mathrm{Tr}(\widehat{A}_c(P)\rho_x). \tag{17}$$

For every pure test state,

$$\left|\mathrm{Tr}\left((A_c - \widehat{A}_c)\rho_x\right)\right| \leq \|O_c\|_2 \leq \|O_c\|_F. \tag{18}$$

The same residual thus bounds both measurement compression and the error of the spectral class-affinity approximation.

6 Monotone Multiclass Jacobi-DECA

For noncommuting classes we maximize the piecewise-smooth objective

$$F(P) = \sum_j \max_c p_j^\dagger A_c p_j. \tag{19}$$

Given a hard assignment z_j, consider components p_j, p_k assigned to $c = z_j$ and $r = z_k$. If $c \neq r$, set

$$Q = [p_j, p_k], \qquad \widehat{B} = Q^\dagger (A_c - A_r) Q. \tag{20}$$

The optimal updated first direction in this two-dimensional subspace is the principal eigenvector of the Hermitian 2×2 matrix $\widehat{B}$; the second is its orthogonal complement. If $c = r$, the pair contribution is rotation-invariant.

Proposition 7 (Monotonicity)**.** *Every assignment step and pair rotation in Algorithm 1 is nondecreasing in its corresponding block objective. Since $F(P) \leq 1$, the objective values converge.*

Proof. The assignment step is exact by Theorem 2. For fixed labels, the variable part of a pair objective is the Rayleigh quotient $v^\dagger \widehat{B} v$, maximized by the principal eigenvector. Bounded monotone sequences converge. This does not imply convergence to the global maximizer. □

Constructing empirical class operators costs $O(nd^2)$ time and $O(Kd^2)$ storage. Binary training then uses one $O(d^3)$ eigendecomposition. The transparent reference Jacobi implementation uses dense pair updates with worst-case $O(Td^4)$ work per start; cached transformed-operator variants can reduce a sweep toward $O(Kd^3)$. We therefore report sweep budgets and wall time rather than hiding optimization cost.

For real data, an orthogonal basis has $d(d-1)/2$ continuous degrees of freedom. PVM-DECA adds d discrete class assignments; Spectral-DECA adds at most Kd diagonal weights. An unrestricted real symmetric QDA uses $Kd(d+1)/2$ quadratic coefficients before means and offsets, while a generic complex K-effect POVM has order Kd^2 real parameters subject to positivity and completeness. These are structural degrees of freedom, not claims about compressed circuit gate counts.

7 Encodings and Quantum Compilation

7.1 Classical encodings

Amplitude encoding,

$$\phi(x) = x/\|x\|_2, \tag{21}$$

matches the original squared-component construction but discards norm and identifies x with $-x$ at the density-operator level. Its boundaries are homogeneous quadratic forms.

The affine lift

$$\phi_\tau(x) = \frac{[x^\top, \tau]^\top}{\sqrt{\|x\|_2^2 + \tau^2}} \tag{22}$$

adds linear and bias-like terms through the off-diagonal blocks of $\phi_\tau(x)\phi_\tau(x)^\dagger$. It addresses sign and mean information but introduces a scale τ that must be chosen without test leakage. Stereographic encodings are established prior art in quantum-centroid classification [10].

7.2 Ancilla-free PVM

Pad d to $D = 2^{\lceil \log_2 d \rceil}$, prepare $|\phi(x)\rangle$, apply $P^\dagger$, and measure in the computational basis:

$$\Pr(J = j \mid x) = |p_j^\dagger \phi(x)|^2. \tag{23}$$

The decoder maps j to z_j. No outcome ancilla is required. In the binary case this exactly compiles the Helstrom PVM; Spectral-DECA reuses the same outcomes with real weights and repeated shots.

7.3 General POVM by Naimark dilation

For effects $\{E_c\}$, define $M_c = \sqrt{E_c}$ and the stacked isometry

$$V = [M_1^\top, \dots, M_K^\top]^\top, \qquad V^\dagger V = \sum_c E_c = I. \tag{24}$$

Complete V to a unitary on the enlarged space, initialize an outcome register, and measure that register. A generic flat K-outcome realization requires $\lceil \log_2 K \rceil$ outcome qubits and a substantially larger arbitrary unitary [19]. This is a compilation statement, not a gate-count advantage claim.

8 Experimental Design

All code, tests, exact commands, raw fold records, figures, and software versions are included in the accompanying repository. We used Python 3.10.13, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.7.2, CVXPY 1.7.5 [20], Qiskit 2.5.1 [21], and Qiskit Aer 0.17.2. Seventeen unit tests cover encodings, POVM validity, analytical identities, monotonicity, scikit-learn behavior, and circuit probabilities.

8.1 Theory and circuit experiments

We sampled random real density operators at dimensions $2, 4, 8$ for binary exactness; commuting three-class ensembles at dimensions $4, 8$; and 72 controlled noncommuting three-class ensembles. Every DECA result was compared to the SDP oracle in Equation (4). A trine qubit ensemble provides a standard case where a three-effect POVM outperforms any two-outcome qubit PVM.

Qiskit Aer experiments use 16,384 shots per input state. We simulate (i) a binary, ancilla-free PVM and (ii) an SDP-derived trine POVM through explicit Naimark dilation. Analytic Born probabilities are compared with shot frequencies using total-variation distance.

8.2 Classical protocol

Two synthetic binary datasets isolate zero-mean covariance difference and equal-covariance mean difference. Eight public datasets cover 150 to 20,000 samples, 4 to 64 features, and 2 to 26 classes: Iris, Wine, Breast Cancer, Digits, Banknote Authentication, Spambase, Dry Bean, and Letter Recognition. The UCI archives are downloaded from fixed URLs and verified by SHA-256 [22–26].

We compare logistic regression, shrinkage LDA, regularized QDA, degree-two and RBF SVMs [27], random forests [28], histogram gradient boosting, PVM-DECA, Spectral-DECA, and PGM. Standardization is fitted inside each training fold. Affine scale is fixed to $\sqrt{m}$ for m input features. Jacobi budgets are deterministic and dimension-dependent: 10 sweeps and no random starts for $m \geq 32$, 20 sweeps for $12 \leq m < 32$, and 30 sweeps plus two random starts below 12 dimensions.

We use two repeats of five-fold stratified cross-validation. The study uses fixed, non-nested configurations and must therefore be read as a mechanism and resource comparison, not a state-of-the-art hyperparameter contest. Accuracy, balanced accuracy, macro-F1, single-shot expected success, fit time, prediction time, Jacobi sweeps, and residuals are saved for all 1,100 outer fits. Dataset-level mean accuracies

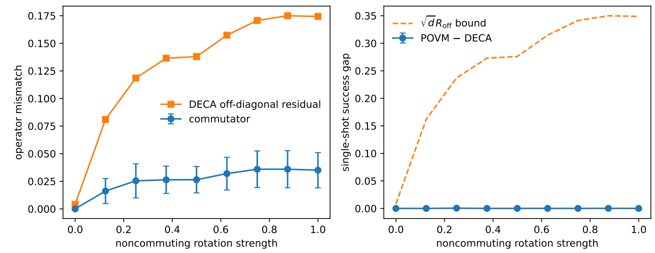

Figure 1: Controlled noncommuting ensembles. The off-diagonal bound is valid but loose; rotation strength and commutator norm do not imply a monotone oracle gap. Error bars show variation across eight seeds.

are independent blocks in the Friedman and exploratory Wilcoxon tests [29]; overlapping folds are never treated as independent observations.

9 Results

9.1 Theorem validation

Across 30 binary trials, the largest direct closed-form identity error was 8.9×10^{-16} and the largest discrepancy from the SDP oracle was 2.0×10^{-8}. Across 16 commuting multiclass trials, the largest SDP gap was 1.45×10^{-8}. None of 72 noncommuting trials violated Equation (13). The empirical Pearson correlation between commutator norm and actual gap was -0.11, so noncommutativity alone is not a calibrated performance predictor; the basis-specific residual provides the proved certificate. Figure 1 shows the sweep.

9.2 Circuit validation and POVM advantage

The maximum total-variation error between Qiskit shot frequencies and analytic probabilities was 0.0073. Binary PVM-DECA achieved analytic training single-shot success 0.9434 with no outcome ancilla. On the equal-prior trine ensemble, the best commuting PVM found by Jacobi-DECA achieved 0.6220, while PGM and the SDP POVM achieved 2/3, an absolute advantage 0.04466. The general realization used two padded outcome qubits. Figure 2 reports analytic and sampled success.

9.3 Classical mechanism and predictive accuracy

Table 1 reports repeated-CV accuracy. The covariance-only task demonstrates the intended mechanism: logistic regression is at 0.496 ± 0.019, whereas Spectral-DECA-amplitude reaches 0.786 ± 0.014 (shown in the complete table in Table 3). PVM-DECA reaches only 0.623 ± 0.038 because sign-only Helstrom weights solve a different single-shot objective. On the mean-only task, amplitude variants

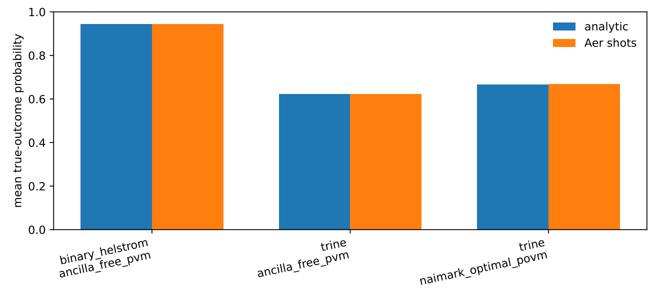

Figure 2: Analytical and 16,384-shot Qiskit Aer success. The trine ensemble exhibits a real POVM advantage; the binary PVM is ancilla-free and exact.

are near chance, while the affine spectral rule reaches 0.825 ± 0.017, matching logistic regression.

The public-data conclusion is deliberately conservative. PGM is competitive on Wine (0.980) and trails strong baselines moderately on Breast Cancer (0.950) and Digits (0.941), but PVM and spectral variants are generally weaker. Across the eight public datasets, the Friedman statistic is 57.44 with $p = 1.10 \times 10^{-8}$. Dataset-level exploratory Wilcoxon tests with Holm correction give:

- PGM minus PVM: mean $+0.202$, seven wins in eight datasets, adjusted $p = 0.0469$;
- PGM minus RBF-SVM: mean -0.0886, one win in eight, adjusted $p = 0.0469$.

Only eight independent dataset blocks are available, so these pairwise tests are supporting evidence rather than definitive population-level rankings.

Letter Recognition supplies a sharp structural stress test: $K = 26 > d = 17$ after affine lifting. In agreement with Proposition 6, PVM-DECA achieves 0.311 ± 0.010; a 26-effect PGM improves to 0.617 ± 0.003; RBF-SVM reaches 0.971 ± 0.002. Dry Bean, where $K < d$ but the shared-basis assumption remains restrictive, similarly separates PVM (0.616), PGM (0.829), and RBF-SVM (0.931).

9.4 Resource observations

Table 2 shows median wall-clock measurements. Binary DECA fits are analytical and often millisecond-scale. Multiclass Jacobi cost grows with dimension and sweep budget: on Digits, PVM training takes 2.12 s, versus 0.034 s for the fixed RBF-SVM configuration and 0.15 s for PGM. On large, low-dimensional Letter, PVM and PGM prediction are faster per sample than RBF-SVM but sacrifice large amounts of accuracy. These software timings are not hardware, power, or asymptotic quantum advantage measurements.

10 Applications and Design Guidance

Native quantum data. The strongest use case is measurement design when inputs already are quantum states: qubit readout, communication receivers, sensing, calibration, and drift monitoring. There is no classical state-loading cost. Binary and commuting ensembles admit exact ancilla-free measurements; the residual and SDP gap quantify when a more expensive POVM is justified.

Classical second-order signals. Spectral-DECA is most plausible for zero-mean or covariance-dominated measurements: EEG/BCI, radar and RF, vibration faults, texture or scattering features, and few-shot normalized embeddings. The covariance-only experiment supports this mechanism. A practitioner should compare class-operator commutators, fit a common basis, and inspect R_{off} before assuming that the shared-basis bias is appropriate.

Decision guide. Use binary closed-form DECA when a single physical measurement is the true loss. Use Spectral-DECA when eigenvalue magnitude is meaningful and repeated expectation estimation is available. Use PGM when multiclass accuracy justifies nonprojective effects and a closed form is preferred. Use the SDP oracle for small validation problems. If $K > d$, enlarge the encoding, use a general POVM/Naimark dilation, or use hierarchical measurements. For ordinary tabular prediction, strong classical baselines remain mandatory.

11 Limitations and Reproducibility

First, ρ_c retains only the class average of encoded rank-one operators. Higher-order and multimodal within-class information can be lost. Second, amplitude normalization discards norm; affine lifting introduces a scale that was fixed rather than nested-tuned here. Third, the Jacobi objective is nonconvex. Monotonicity guarantees objective convergence, not global optimality or test-accuracy improvement. Fourth, the bound in Equation (13) can be loose and does not reverse: a small actual gap does not imply a small residual. Fifth, repeated-CV standard deviations are descriptive because folds overlap; the independent-block tests use only eight public datasets. Sixth, arbitrary state preparation and dense unitaries can dominate a classical-to-quantum pipeline. We make no end-to-end speedup claim.

The code records all seeds, versions, URLs, archive hashes, fold predictions, measurement diagnostics, and generated figures. Qiskit simulations are local and require no cloud account. Authorship and venue metadata in this draft are placeholders and must be confirmed before submission.

Table 1: Accuracy (mean ± standard deviation over two repeats of five-fold stratified CV). A: amplitude encoding; F: affine encoding. Configurations are fixed rather than nested-tuned.

Dataset	Logistic	QDA	RBF-SVM	RF	PVM-A	PGM-F
Covariance signal	0.496 ± 0.019	0.823 ± 0.022	0.759 ± 0.017	0.824 ± 0.020	0.623 ± 0.038	0.759 ± 0.027
Mean signal	0.825 ± 0.019	0.822 ± 0.016	0.775 ± 0.019	0.828 ± 0.018	0.502 ± 0.042	0.826 ± 0.016
Iris	0.957 ± 0.027	0.957 ± 0.035	0.960 ± 0.034	0.943 ± 0.045	0.543 ± 0.085	0.887 ± 0.050
Wine	0.980 ± 0.023	0.997 ± 0.009	0.978 ± 0.022	0.980 ± 0.023	0.786 ± 0.055	0.980 ± 0.030
Breast Cancer	0.977 ± 0.014	0.965 ± 0.025	0.967 ± 0.029	0.954 ± 0.026	0.675 ± 0.026	0.950 ± 0.026
Digits	0.971 ± 0.007	0.979 ± 0.005	0.982 ± 0.005	0.977 ± 0.008	0.890 ± 0.016	0.941 ± 0.015
Banknote	0.982 ± 0.008	0.981 ± 0.008	1.000 ± 0.000	0.993 ± 0.004	0.708 ± 0.026	0.956 ± 0.011
Spambase	0.926 ± 0.011	0.761 ± 0.008	0.933 ± 0.009	0.954 ± 0.005	0.866 ± 0.008	0.854 ± 0.011
Dry Bean	0.925 ± 0.006	0.920 ± 0.005	0.931 ± 0.004	0.925 ± 0.004	0.616 ± 0.004	0.829 ± 0.005
Letter	0.773 ± 0.005	0.849 ± 0.004	0.971 ± 0.002	0.966 ± 0.003	0.311 ± 0.010	0.617 ± 0.003

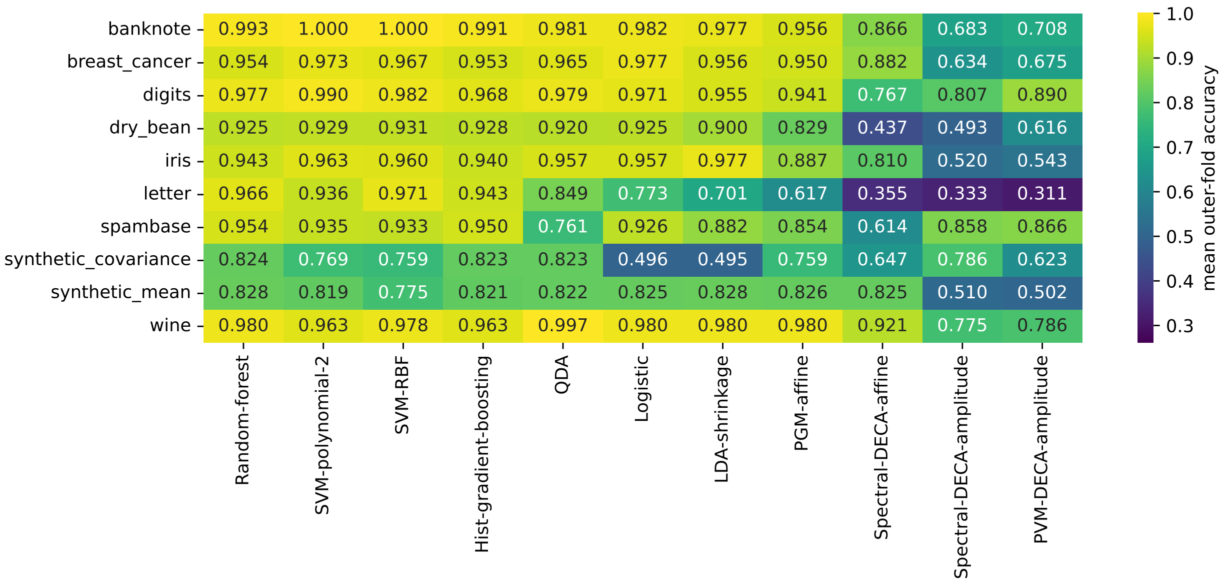

Figure 3: Complete mean-accuracy matrix. The controlled datasets explain *when* energy contrast works; the public datasets expose the substantial cost of a shared measurement basis.

12 Conclusion

DECA gives a precise answer to the original "class difference rather than variance" intuition. The right mathematical object is the spectrum of class-conditional energy operators, and the right physical interpretation is an optimal measurement only after its constraints are stated. A shared unitary is useless before an unrestricted POVM, but valuable as a compressed, interpretable measurement basis. Binary and commuting cases are exact; noncommuting multiclass cases have a computable residual bound and a monotone Jacobi approximation. Retaining eigenvalue magnitudes yields a distinct spectral quadratic classifier.

The experiments support the theory more strongly than a universal performance story: DECA succeeds on controlled second-order structure, general POVMs consistently repair part of the multiclass loss, and ordinary kernel or ensemble methods remain stronger on most tabular tasks. This honest boundary turns ECA from a broad quantum analogy into a falsifiable framework for discriminative components, measurement compression, and resource-aware classification.

A Additional Derivations

A.1 Binary basis objective

For scalars u, v, $\max(u, v) = (u+v+|u-v|)/2$. Therefore

$$\sum_j \max(p_j^\dagger A_1 p_j, p_j^\dagger A_2 p_j) = \frac{1}{2}\sum_j p_j^\dagger (A_1 + A_2) p_j + \frac{1}{2}\sum_j |p_j^\dagger \Delta p_j| = \frac{1}{2}\left[1 + \|\operatorname{diag}(P^\dagger \Delta P)\|_1\right].$$

Table 2: Median outer-fold software resource measurements.

Dataset	Method	Accuracy	Fit (s)	Predict (μs/sample)
Digits	RBF-SVM	0.982	0.03433	34.16
	PVM-A	0.890	2.123	77.92
	PGM-F	0.941	0.15	80.38
Spambase	RBF-SVM	0.933	0.1485	32.08
	PVM-A	0.866	0.06704	16.87
	PGM-F	0.854	0.06983	14.92
Dry Bean	RBF-SVM	0.931	0.214	52.96
	PVM-A	0.616	0.1709	4.094
	PGM-F	0.829	0.06104	4.718
Letter	RBF-SVM	0.971	0.9645	221.4
	PVM-A	0.311	0.4181	14.06
	PGM-F	0.617	0.3396	17.84

This yields Equation (9).

A.2 Spectral score as a structured quadratic model

For real affine encoding before its normalization, let $\widetilde{x} = [x^\top, \tau]^\top$. If $\Delta = P\,\mathrm{diag}(\lambda)P^\top$, then

$$\widetilde{x}^\top \Delta \widetilde{x} = \sum_j \lambda_j (p_j^\top \widetilde{x})^2.$$

Partitioning Δ as $\begin{bmatrix} Q & q \\ q^\top & q_0 \end{bmatrix}$ gives $x^\top Q x + 2\tau q^\top x + \tau^2 q_0$. Spectral-DECA is therefore linear in learned energy features but quadratic, with possible linear and bias terms, in the original input.

B Complete Accuracy Table

C Repository Commands

```
python -m pytest experiments/run1/tests -q
cd experiments
python scripts/run_theory_validation.py
python scripts/run_quantum_simulation.py
python scripts/run_classical_benchmarks.py \
  --folds 5 --repeats 2
python scripts/analyze_classical_results.py
python scripts/export_paper_tables.py
cd ..
make -C publication
```

References

[1] Ronald A. Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936. doi: 10.1111/j.1469-1809.1936.tb02137.x.

[2] Jerome H. Friedman. Regularized discriminant analysis. *Journal of the American Statistical Association*, 84(405):165–175, 1989. doi: 10.1080/01621459.1989.10478752.

[3] Bernhard N. Flury. Common principal components in k groups. *Journal of the American Statistical Association*, 79(388):892–898, 1984. doi: 10.1080/01621459.1984.10477108.

[4] Herbert Ramoser, Johannes Müller-Gerking, and Gert Pfurtscheller. Optimal spatial filtering of single trial EEG during imagined hand movement. *IEEE Transactions on Rehabilitation Engineering*, 8(4):441–446, 2000. doi: 10.1109/86.895946.

[5] Rongzhou Chen. Eigen component analysis: A quantum theory incorporated machine learning technique to find linearly maximum separable components, 2020. URL https://arxiv.org/abs/2003.10199.

[6] Rongzhou Chen, Yaping Zhao, Hanghang Liu, Haohan Xu, Shaohua Ma, and Edmund Y. Lam. Eigen-component analysis: A quantum theory-inspired linear model. In *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–5, 2025. doi: 10.1109/ISCAS56072.2025.11044249.

[7] Carl W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, New York, 1976.

[8] Yonina C. Eldar, Alexandre Megretski, and George C. Verghese. Designing optimal quantum detectors via semidefinite programming. *IEEE Transactions on Information Theory*, 49(4):1007–1012, 2003. doi: 10.1109/TIT.2003.809510.

[9] Paul Hausladen and William K. Wootters. A "pretty good" measurement for distinguishing quantum states. *Journal of Modern Optics*, 41(12):2385–2390, 1994. doi: 10.1080/09500349414552221.

[10] Giuseppe Sergioli, Roberto Giuntini, and Hector Freytes. A new quantum approach to binary classification. *PLOS ONE*, 14(5):e0216224, 2019. doi: 10.1371/journal.pone.0216224.

[11] Roberto Giuntini, Hector Freytes, Daniel K. Park, Carsten Blank, Federico Holik, Keng Loon Chow, and Giuseppe Sergioli. Quantum state discrimination for supervised classification, 2021. URL https://arxiv.org/abs/2104.00971.

[12] Seth Lloyd, Maria Schuld, Aroosa Ijaz, Josh Izaac, and Nathan Killoran. Quantum embeddings for machine learning, 2020. URL https://arxiv.org/abs/2001.03622.

[13] Emmanuel Zambrini Cruzeiro, Christine De Mol, Serge Massar, and Stefano Pironio. Quantum-inspired classification based on quantum state discrimination, 2023. URL https://arxiv.org/abs/2303.15353.

[14] Wooseop Hwang, Daniel K. Park, Israel F. Araujo, and Carsten Blank. Quantum-inspired classification via efficient simulation of helstrom measurement, 2024. URL https://arxiv.org/abs/2403.15308.

Table 3: Mean accuracy for all methods. Spec-A and Spec-F denote spectral DECA with amplitude and affine encodings. HGB denotes histogram gradient boosting.

Dataset	Logistic	LDA	QDA	Poly-SVM	RBF-SVM	RF	HGB	PVM-A	Spec-A	Spec-F	PGM-F
Covariance signal	0.496	0.495	0.823	0.769	0.759	0.824	0.823	0.623	0.786	0.647	0.759
Mean signal	0.825	0.828	0.822	0.819	0.775	0.828	0.821	0.502	0.510	0.825	0.826
Iris	0.957	0.977	0.957	0.963	0.960	0.943	0.940	0.543	0.520	0.810	0.887
Wine	0.980	0.980	0.997	0.963	0.978	0.980	0.963	0.786	0.775	0.921	0.980
Breast Cancer	0.977	0.956	0.965	0.973	0.967	0.954	0.953	0.675	0.634	0.882	0.950
Digits	0.971	0.955	0.979	0.990	0.982	0.977	0.968	0.890	0.807	0.767	0.941
Banknote	0.982	0.977	0.981	1.000	1.000	0.993	0.991	0.708	0.683	0.866	0.956
Spambase	0.926	0.882	0.761	0.935	0.933	0.954	0.950	0.866	0.858	0.614	0.854
Dry Bean	0.925	0.900	0.920	0.929	0.931	0.925	0.928	0.616	0.493	0.437	0.829
Letter	0.773	0.701	0.849	0.936	0.971	0.966	0.943	0.311	0.333	0.355	0.617

[15] Angelika Bunse-Gerstner, Ralph Byers, and Volker Mehrmann. Numerical methods for simultaneous diagonalization. *SIAM Journal on Matrix Analysis and Applications*, 14(4):927–949, 1993. doi: 10.1137/0614062.

[16] Jean-François Cardoso and Antoine Souloumiac. Jacobi angles for simultaneous diagonalization. *SIAM Journal on Matrix Analysis and Applications*, 17(1):161–164, 1996. doi: 10.1137/S0895479893259546.

[17] Rajendra Bhatia. *Matrix Analysis*. Springer, New York, 1997. doi: 10.1007/978-1-4612-0653-8.

[18] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

[19] Mark A. Naimark. Spectral functions of a symmetric operator. *Izvestiya Rossiiskoi Akademii Nauk. Seriya Matematicheskaya*, 4(3):277–318, 1940.

[20] Steven Diamond and Stephen Boyd. CVXPY: A python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

[21] Ali Javadi-Abhari, Matthew Treinish, Kevin Krsulich, Christopher J. Wood, Jake Lishman, Julien Gacon, Simon Martiel, Paul D. Nation, Lev S. Bishop, Andrew W. Cross, Blake R. Johnson, and Jay M. Gambetta. Quantum computing with Qiskit, 2024. URL https://arxiv.org/abs/2405.08810.

[22] Dheeru Dua and Casey Graff. UCI machine learning repository. University of California, Irvine, School of Information and Computer Sciences, 2019. URL https://archive.ics.uci.edu/.

[23] Volker Lohweg. Banknote authentication, 2012.

[24] Mark Hopkins, Erik Reeber, George Forman, and Jaap Suermondt. Spambase, 1999.

[25] Dry bean, 2020.

[26] David Slate. Letter recognition, 1991.

[27] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine Learning*, 20:273–297, 1995. doi: 10.1007/BF00994018.

[28] Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.

[29] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.